Inequalities — problem 7.71

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Let $x_i > 0$, $x_1 + \cdots + x_n = 1$, $n \geq 2$. Prove that $S \geq n/(2n-1)$ if

$$S = \frac{x_1}{1 + x_2 + \cdots + x_n} + \frac{x_2}{1 + x_1 + x_3 + \cdots + x_n} + \cdots + \frac{x_n}{1 + x_1 + \cdots + x_{n-1}}.$$